OSIRIS QED Extension: Axion Compton Module (Thermal Bath)

Summary of implementation behavior and algorithms. Generated 2026-02-02.

1. Physical process modeled

The Compton channel implemented here models axion production via thermal-photon photoproduction on electrons:

```
γ + e  →  a + e
```

This is treated in a thermal-bath approximation: the photon population is not represented as explicit photon macroparticles. Instead, a local temperature T defines a blackbody photon bath that drives axion emission. This mirrors the structure used in the thermal axion-bremsstrahlung module: emission is an effective stochastic source term applied to electrons.

2. Local emissivity and per-electron rate

The code evaluates Raffelt's thermally averaged Compton energy-loss rate per unit mass (Phys. Rev. D 33, 897 (1986), Eq. 36), which depends on temperature, composition (via hydrogen mass fraction X_H), and the electron coupling α_ae = g_ae^2/(4π).

In each cell, the module forms a volumetric emissivity (power density) and converts it to a per-electron emission rate:

```
ε_C(T, X_H, g_ae)          [energy / mass / time]
Q_C = ρ * ε_C              [energy / volume / time]
Γ_C ≈ Q_C / (n_e * E_mean) [1 / time per electron]
```

Here ρ is the local mass density and n_e is the local electron density. To remain consistent with the bremsstrahlung implementation choice (your option A), ρ is obtained from the deposited ion density and n_e is inferred from quasi-neutrality using the configured ion charge Z_ion.

E_mean is the mean axion energy for the thermal Compton spectrum (order kT). The code's sampling kernel is consistent with Raffelt's reported spectral characteristics (peak and mean in units of E/T).

3. Emission operator (Poisson + unbiased variance control)

For each electron macroparticle, the expected number of emitted macro-axions in one timestep Δt is computed. Emission is then decided using a Poisson draw, matching the bremsstrahlung method.

```
λ_macro  ∝  |q_e| * Γ_C * Δt / w_a
k_emit ~ Poisson(λ_eff)
```

w_a is the macro-axion weight (set by input parameter w_axion, or derived from a multiplicity factor proba_mult in the legacy mode). To reduce variance while keeping the estimator unbiased, a cap λ_cap may be applied:

```
if λ_macro > λ_cap:
    λ_eff = λ_cap
    w_a   = w_a * (λ_macro / λ_cap)   (weight rescaling preserves ⟨Q⟩)
```

This cap + weight rescaling keeps the expected emissivity unchanged while preventing rare particles from spawning an excessively large number of macro-axions in a single step.

4. Sampling emitted axion kinematics

If k_emit > 0, the module creates that many axion macroparticles. For each emitted axion:

```
1) Sample energy from the thermal Compton spectral shape.
   (For the power spectrum in this approximation: dQ/dE ∝ E^5 / (exp(E/T) - 1).)
2) Convert sampled energy to momentum magnitude (massless axion assumption in the PIC particle
3) Sample an isotropic emission direction and construct the momentum vector.
4) Create the axion particle with weight w_a.
```

The isotropic direction sampling matches the default used in the thermal bremsstrahlung module; the thermal-bath model does not attempt to resolve photon angular anisotropy or explicit electron recoil distributions.

5. Plasma feedback (optional)

When apply_feedback is enabled, the module deposits the emitted axion energy and momentum back into diagnostic/feedback buffers using the same pattern as bremsstrahlung. This allows the electron population (or your chosen energy bookkeeping path) to lose energy/momentum consistent with the emitted axions.

6. Inverse process (absorption) via detailed balance

The inverse channel (a + e → γ + e) is implemented as a stochastic absorption/removal operator on existing axion macroparticles. For each axion particle of energy E in a cell, the module computes an absorption rate Γ_abs(E) such that the forward and inverse operators satisfy detailed balance with the same thermal bath and the same forward spectral emissivity model.

```
Construct forward spectral rate:   d(dot n_a)/dE
Construct equilibrium spectrum:    dn_a^eq/dE  (thermal bath)
Set absorption rate:               Γ_abs(E) = [d(dot n_a)/dE] / [dn_a^eq/dE]
Absorb with probability:           P_abs = 1 - exp(-Γ_abs(E) * Δt)
```

If an axion is absorbed, it is removed from the axion species. If feedback is enabled, the absorbed axion energy/momentum is deposited with the opposite sign of emission (consistent bookkeeping).

7. Input parameters

The Compton block mirrors the bremsstrahlung configuration style. Typical parameters include: if_compton and if_inverse; coupling g_ae (or α_ae) and an overall scale cs_scale; thermodynamics and composition (Te_keV or evolving Te, X_H, A_ion); ion linkage for the density proxy (Z_ion, i_ion(:)); variance controls (w_axion, proba_mult, lambda_cap); and optional feedback controls (apply_feedback, fe_electrons, and related buffers).

Overall, the Compton module follows the same algorithmic structure as the thermal axion-bremsstrahlung implementation: cell-local emissivity → per-electron rate → Poisson emission of weighted macro-axions, plus an optional inverse operator based on detailed balance.